# 9.6 Solving Right Triangles

Name _____ Period ____ Date _____

Geometry 9.6 (30-minute plan) · Student sheet

**I can…**

- use an inverse trigonometric ratio to find an angle measure.
- solve a right triangle by finding all of its side lengths and angle measures.
- solve real-life problems with right triangles.

**Vocabulary** · fill in the missing word

**Inverse tangent** — if tan A = _____, then $\tan^{-1} x = m\angle A$

**Inverse sine and inverse cosine** — $\sin^{-1}$ and $\cos^{-1}$: they give the angle _____ from a sine or cosine ratio

**Solve a right triangle** — find all of the unknown side lengths and angle _____

## Part 1 · Notes and examples (fill in each note, then work the example)

**Inverse trig ratios:** if tan A = x, then _____ = m∠A. It works the same way with $\sin^{-1}$ and _____.

### Example 1

Find m∠A to the nearest tenth of a degree.

a. tan A = 0.75

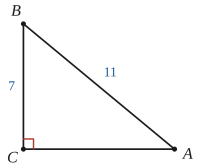
b. sin A = 0.5

c. cos A = 0.8

**Which ratio?** Use the two sides you _____: opposite and adjacent → tan, opposite and hypotenuse → _____, adjacent and hypotenuse → _____.

### Example 2

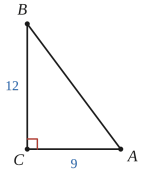
Find m∠A to the nearest tenth of a degree.




**To solve a right triangle:** find the third side with the _____ Theorem, one acute angle with an inverse ratio, and the other acute angle with 90° minus the first.

### Example 3

Solve right △ABC: find AB, m∠A, and m∠B. Round to the nearest tenth.



## Part 2 · Partner practice

### Partner practice

In right △ABC, ∠C is the right angle, AC = 8, and BC = 5. Find AB, m∠A, and m∠B. Round to the nearest tenth.

